

THE IONOSPHERIC

The first major use of the ionosphere was through early radio experiments. In 1901 Guglielmo Marconi sent a radio signal across the Atlantic Ocean. Scientists originally believed radio waves only travelled in straight lines, so this experiment proved that signals were reflecting from an invisible layer in the atmosphere.

As communication technology developed, humans became more dependent on the ionosphere. High frequency radio systems used the atmosphere like a mirror, reflecting signals across distances.



INTERSECTION



A study in my project is the Duga radar near the Chernobyl Nuclear Power Plant. Duga was a Soviet early warning system designed to detect missile launches from thousands of kilometres away. It worked by sending radio signals into the ionosphere and analysing the signals that returned to Earth. I find Duga interesting because it was physically massive. The structure was around 150 metres high and hidden in forests near Pripyat.

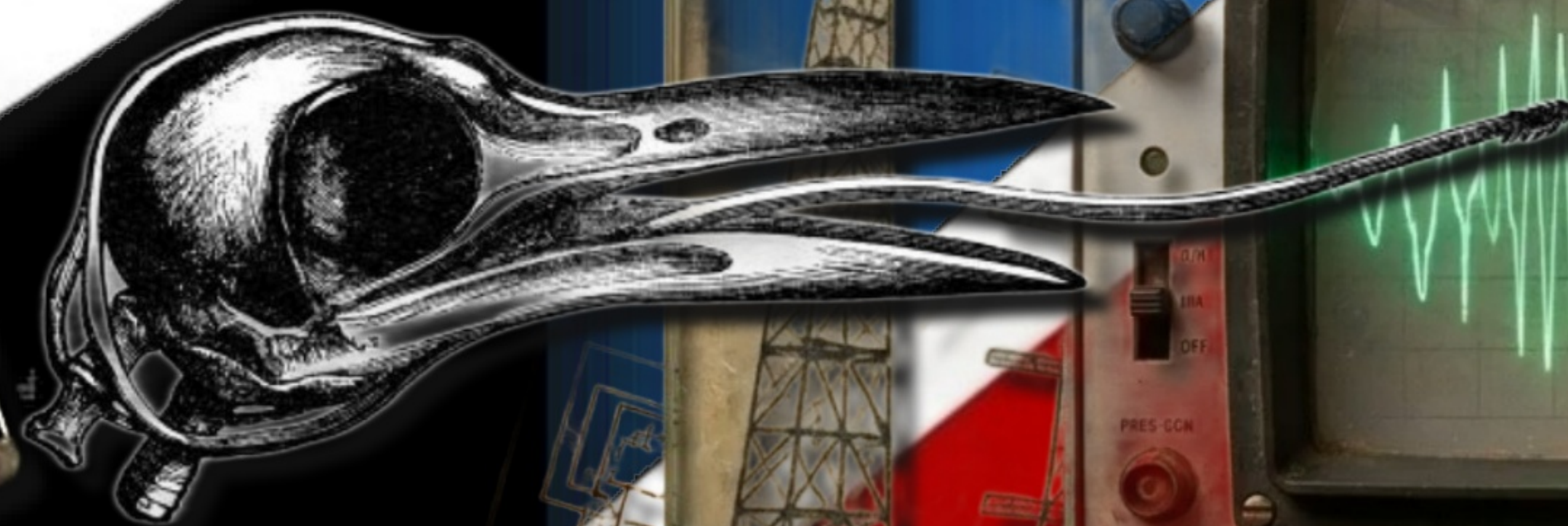




glitches symbolise:

damaged memory
censorship
broken archives
signal interference
unreliable media systems
fragmented historical truth

Sound is also important in the project. The repetitive Woodpecker tapping creates feelings of anxiety, surveillance, and tension. the Russian Woodpecker Hunting Club was an Amateur radio operators network in America. They tried to disrupt the Soviet radar by sending tapping signals back at it. symbolically because civilians were fighting invisible military technology through interference and counter signals it was surprising that their method of tapping the microphone back at the signal actually made the radar breakdown for long periods.

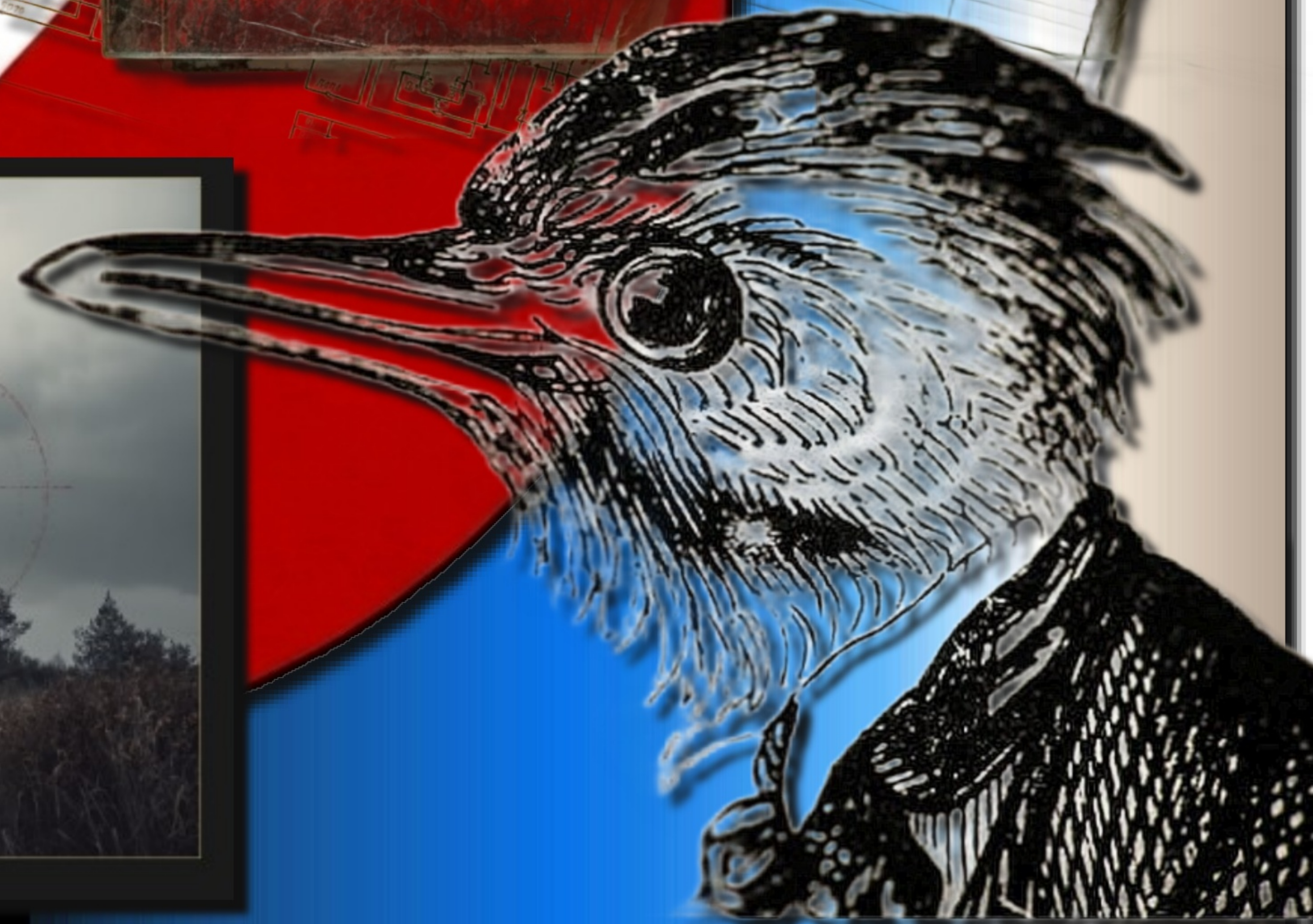


The impact of Duga's radar technology was extensive

100,000 km²

Area affected by Duga's radar signals

Duga's unique radar system influenced a **vast region**, enabling surveillance capabilities across a significant part of the ionosphere and altering communication landscapes.



Invisible Fear

The mysterious tapping sound instilled **anxiety** about unseen technological threats affecting society.

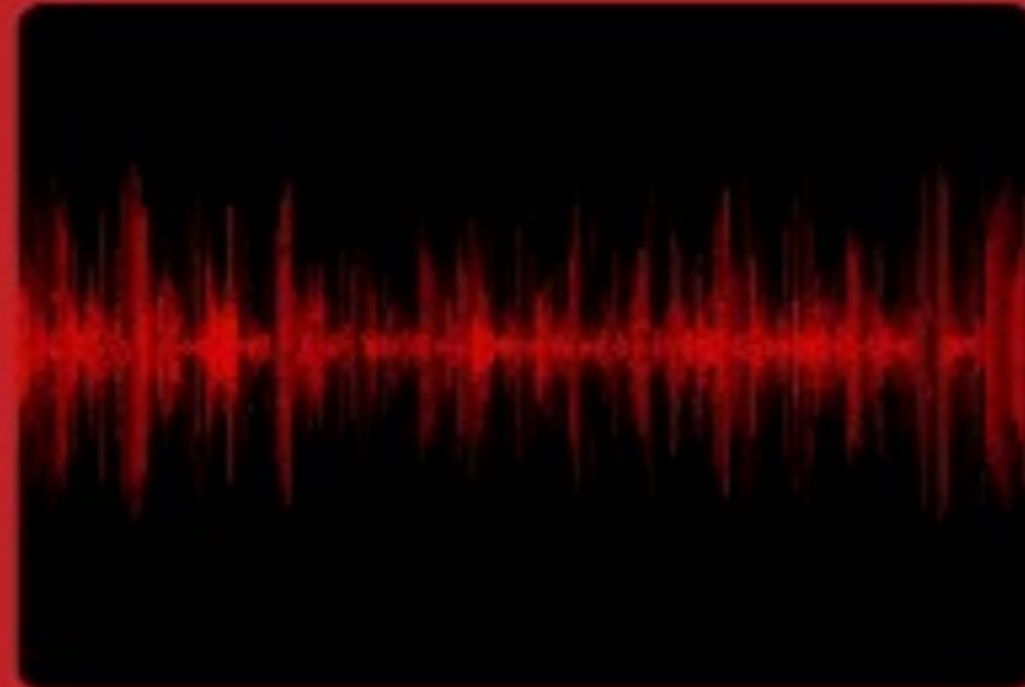


Because the Soviet Union never properly explained the source of the signal, many conspiracy theories appeared.

Some people believed the signal was:
a mind control weapon
a weather control device
a psychological warfare system

Signal Disruption

The radio signal disrupted global communications, causing **unpredictable interference** across various systems.



This idea connects strongly to the documentary *The Russian Woodpecker* by Fedor Alexandrovich. The film presents the theory that the Chernobyl disaster was used to hide failures in the Duga programme. There is no scientific evidence proving this, but the theory itself is powerful because it shows how political secrecy can create paranoia and distrust.

My project is not trying to prove conspiracy theories. Duga used radio signals and could not have caused the nuclear disaster. Instead, I am exploring how invisible systems affect human psychology and how fear grows around technologies people cannot fully understand.



КИЙ
И
СТА

ВО ХИПИ-ЕЛОИ,
ЭТО БУДУЩЕЕ БЕЗГРЕЙНАЯ!

ИСТО ДУН

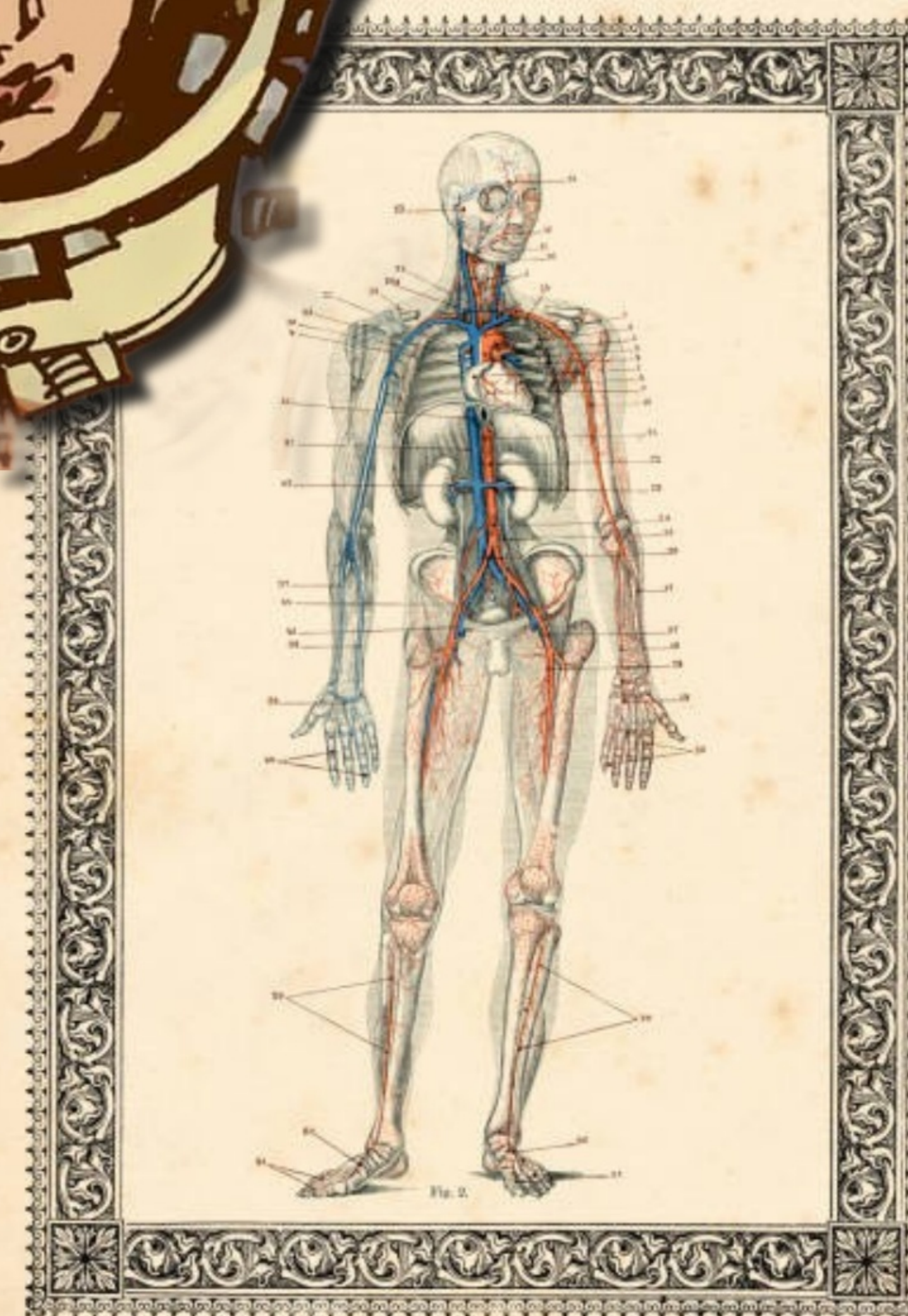
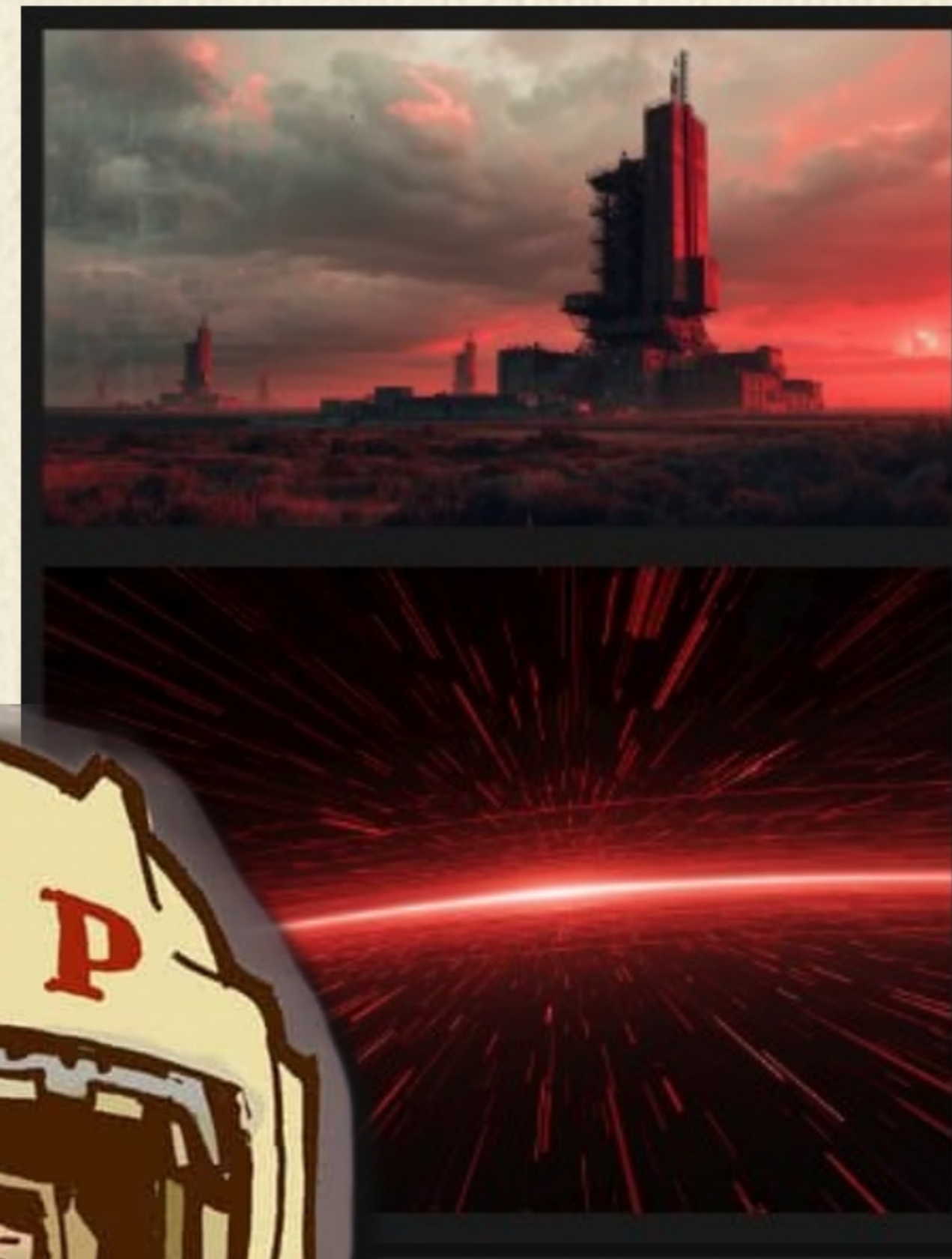
НАЦИОНАЛЬНО УИЛИИ
И КВИНКОМ СУДВИСПАНТОВ
ОГ ДИВЕРСИМ, КРРИКА ПЛОДРЫВАНИИ
И ПРОПОСТАВЕ



ЫН С ДНОЙ
СПИ

ПРОКАЯ
ВОЛУШЬ
В ОАДАВА

ДЕНИЕ-УДНЧЙ



Subversive Signage: Hiding in Plain Sight

The Soviet Union excelled at hiding large military installations through "pathological secrecy".

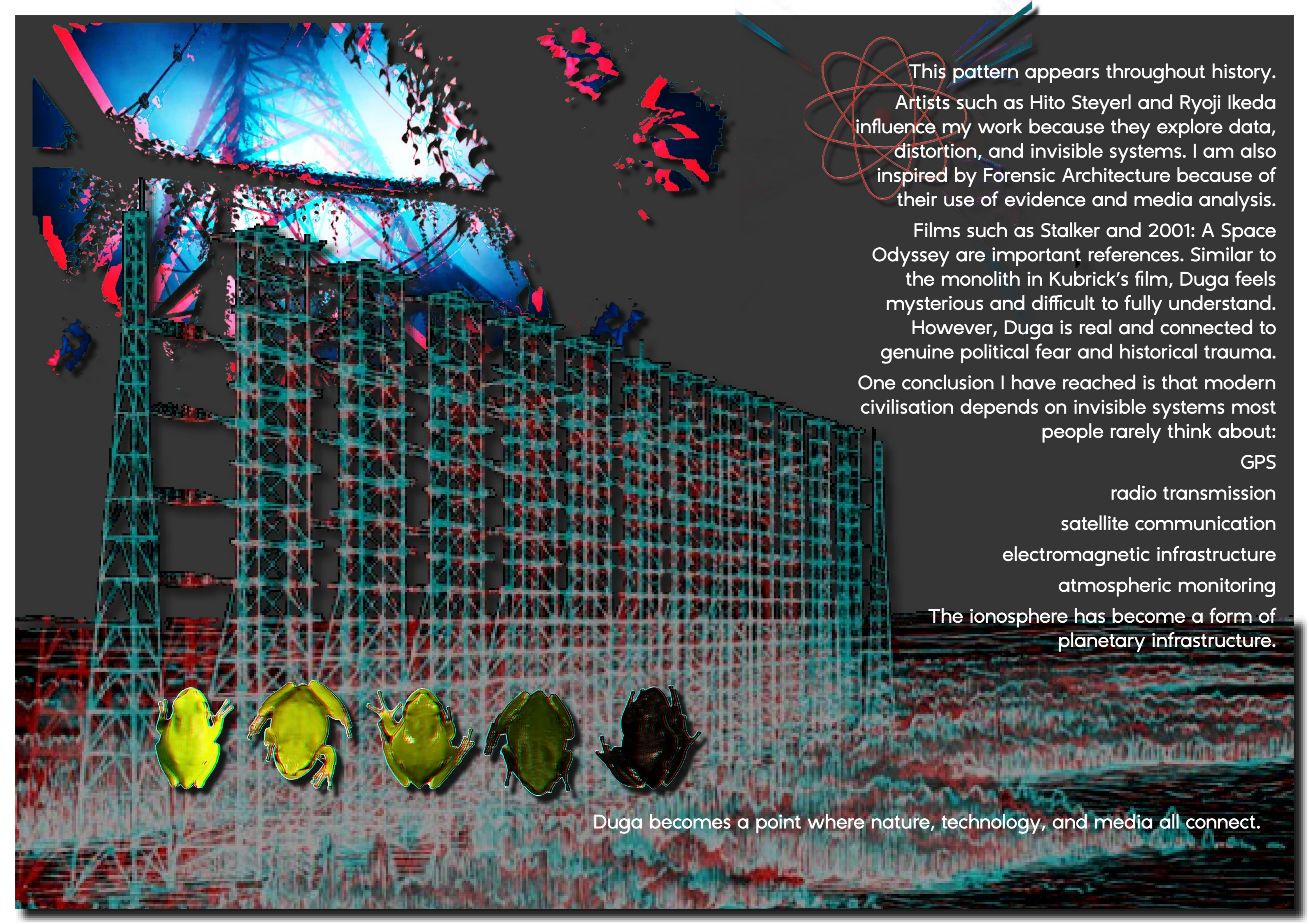
The "Pioneer Camp" Deception:

Maps of the region around Chernobyl-2 intentionally mislabelled the Duga site as a children's camp. We could feature weathered posters for this fictional camp, contrasting innocent imagery with the "brutal, angular" reality of the steel lattice.

Cover Identities: Personnel at the secret city were registered as students of Pripyat or used cover addresses in distant cities to maintain the facility's invisibility. Stencilled administrative codes or forged identity documents would emphasise this forced anonymity.

The "Steel Yard" Designation: While the Soviets called the project Duga (meaning "Arc"), NATO intelligence used the designation "Steel Yard" or "Steel Works". Using these Western military labels in graffiti style creates a visual "dialogue" between opposing intelligence agencies.





This pattern appears throughout history.
Artists such as Hito Steyerl and Ryoji Ikeda influence my work because they explore data, distortion, and invisible systems. I am also inspired by Forensic Architecture because of their use of evidence and media analysis.

Films such as *Stalker* and *2001: A Space Odyssey* are important references. Similar to the monolith in Kubrick's film, Duga feels mysterious and difficult to fully understand.

However, Duga is real and connected to genuine political fear and historical trauma.

One conclusion I have reached is that modern civilisation depends on invisible systems most people rarely think about:

GPS

radio transmission

satellite communication

electromagnetic infrastructure

atmospheric monitoring

The ionosphere has become a form of planetary infrastructure.

Duga becomes a point where nature, technology, and media all connect.

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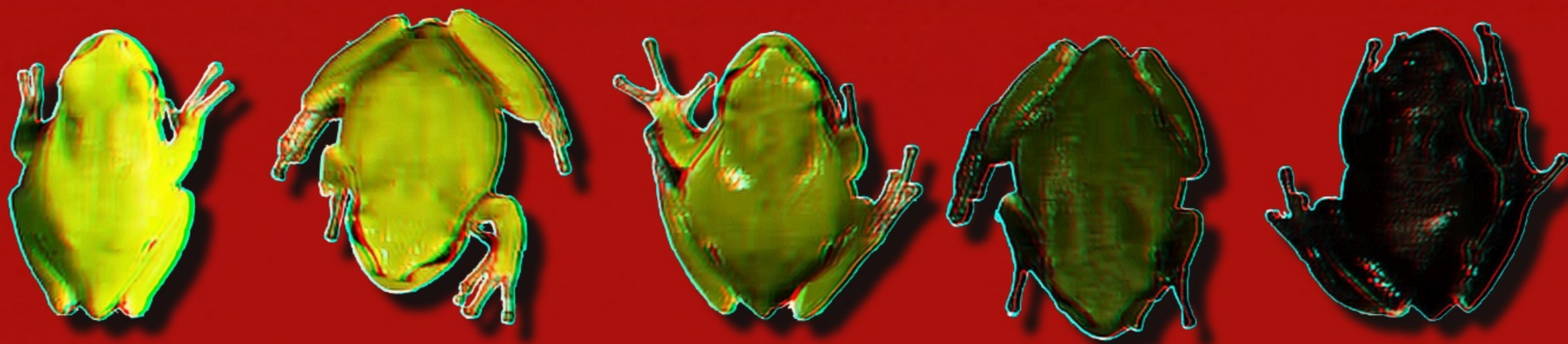
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Is Nature Taking Over?

Altimeter Guide

Space

Ionosphere

Ground Level

Exploring the balance of decay and technological remnants

Overall, I O3 is becoming less about architecture in a traditional sense and more about hidden systems connecting Earth, media, warfare, and space. Duga works both as a historical object and as a symbol of invisible technological power. The ionosphere becomes the central subject of the project because it is an unstable space where nature, communication, surveillance, and human fear all meet.

The Fall of the Giants

The 7-billion-ruble Duga-3 radar was built to master the skies and protect a superpower. Instead, a catastrophic flaw in the very nuclear engineering meant to power it (Chernobyl Reactor 4) rendered the entire region a radioactive ghost town in 1986. The ultimate machine of the Cold War was defeated not by missiles, but by human error and the invisible force of radiation.

First Nature

forest, atmosphere, plasma, ionosphere

Second Nature

radar structures and nuclear infrastructure

Third Nature

signals, media systems, paranoia, and communication networks